7.6.3 Sustainable manufacturing and end-of-life disposal of drones

As drone technology advances and production scales up globally, the focus on sustainable manufacturing and responsible end-of-life disposal has become increasingly important. While drones are often associated with efficiency and environmental monitoring, their own manufacturing processes, material composition, and disposal methods present notable environmental challenges. The rising number of drones in civilian, industrial, and defense sectors means that without sustainable practices, the industry could contribute to resource depletion, electronic waste (e-waste), and pollution, undermining its environmental benefits.

Most commercial drones are built using lightweight composite materials, such as carbon fiber-reinforced polymers (CFRP), thermoplastics, and aluminum alloys. Although these materials provide strength and weight advantages, their production is energy-intensive, and recycling is complex and limited. Carbon fiber, for instance, is difficult to recover once bonded with resins, and improper disposal leads to long-term environmental persistence. Moreover, drones rely heavily on lithium-based batteries, which contain toxic elements like cobalt, nickel, and manganese. If not properly managed, these batteries can leak harmful chemicals into soil and water systems, posing significant ecological risks.

To mitigate these challenges, drone manufacturers and researchers are increasingly adopting sustainable design and production principles. This includes the use of eco-friendly, recyclable materials, such as bio-based composites, biodegradable polymers, and lightweight metals with high recyclability rates. Manufacturing processes are being optimized through additive manufacturing (3D printing), which minimizes material waste and energy consumption while allowing for precision engineering. Some